Brady S. Hardiman

Assistant Professor bhardima@purdue.edu Google Scholar Profile
Purdue University, Forestry and Natural Resources
Environmental and Ecological Engineering
West Lafayette, IN, USA

Education

2012 Ph.D. in Ecology
The Ohio State University, Columbus, OH.
Advisor: Peter Curtis, Dept. of Evolution, Ecology & Organismal Biology
Resilience of Forest Carbon Storage through Disturbance and Succession

2003 Bachelor's of Science, Magna cum Laude, Ashland University.
Comprehensive major in Biology, minor in Chemistry

Professional Experience

Jan. 2016- Assistant Professor, Forestry & Natural Resources and Environmental & Ecological Engineering, Purdue University, West Lafayette, IN

June 2014- Dec. 2015
Post-Doctoral Associate, Carbon Cycle Science & Urban Ecology Laboratory
Dept. of Earth and the Environment, Boston University, Boston, MA
Post-Doctoral Mentor Dr. Lucy Hutyra

June 2012- June 2014
Post-Doctoral Associate**,** Ecological Forecasting Laboratory,
Dept. of Earth and the Environment, Boston University, Boston, MA
Post-Doctoral Mentor Dr. Michael Dietze

2006-2012 PhD Student, The Ohio State University, Columbus, OH.
Dept. of Evolution, Ecology & Organismal Biology

2004-2006 Laboratory Technician**,** Molecular Cytology Core
University of Missouri, Columbia, MO

2003 NSF Research Experience for Undergraduates (REU) Fellow, Harvard Forest, Petersham, MA

Teaching Experience

2016-present FNR 210: Natural Resource Information Management (Spring semesters)
FNR 598: Urban Ecology (Fall Semesters)

2012 Introduction to Biology (Bio 102) Graduate Teaching Assistant**,** Center for Life Sciences Education, The Ohio State University, Columbus, OH.

2011 Evolution (EEOB 400) Graduate Teaching Assistant**,** Dept. of Evolution, Ecology & Organismal Biology (EEOB), The Ohio State University, Columbus, OH.
Introduction to Biology (Bio 102) Graduate Teaching Assistant**,** Center for Life Sciences Education, The Ohio State University, Columbus, OH.

2010 Ecology (EEOB 503) Instructor**,** Dept. of EEOB, The Ohio State University, Columbus, OH.
Honors Evolution (EEOB H400) Graduate Teaching Assistant**,** Dept. of EEOB, The Ohio State University, Columbus, OH.

2009 Theoretical Ecology II (EEOB 702) Graduate Teaching Assistant**,** Dept. of EEOB, The Ohio State University, Columbus, OH.

Peer-Reviewed Publications (IF=Impact Factor)

Dr. Hardiman (h-index=19, i10 index=24) has published 35 peer-reviewed journal articles, 24 of them since 2016. Of Dr. Hardiman's 40 published or in-review publications, 34 (85%) are in the top tier of journals ranked by impact factor (IF = 2.6-14.8) and 5 (13%) are within the middle tier (IF =1.5-2.6). (Tier rankings calculated by combining journals categorized under "Ecology", "Forestry", "Urban Studies", "Regional and Urban Planning", and "Remote Sensing"; N=348 journals; each tier contains 116 journals.) Dr. Hardiman's average publication rate is >4 publications per year since joining Purdue, with a median IF of 4.3 (50% of publications have IFs higher than this). These have been cited 1,403 times, or an average of 40 citations per article.

The lists below indicate the publication year and citation information for each manuscript and the journal impact factor (IF). † indicates advised graduate student author, § indicates advised undergraduate student author, # indicates advised post-doctoral scholar author. A description of Dr. Hardiman's contribution to a publication is provided for articles on which he is not listed as the first author.

Refereed Journal Articles (N=35 as of 9/29/2021)

35. McNickle, G.G., Ritzi, M.V., Blackstone, K.M.S., Couture, J.J., Nelson, T., Hardiman, B.S., Montague, M.S., Jacobs, D.F. Pulling on multiple threads of evidence to reveal tree species coexistence. *Preprint in BioRχiv*. http://doi.org/10.1101/2021.07.13.452199. *In review at Nature Communications.*

    Contribution: Provided feedback on the analysis, edited the manuscript.

34. Francomano, D. Valenzuela, A., Gottesman, B., González-Calderón, A., Anderson, C., Hardiman, B., Pijanowski, B. Acoustic monitoring shows invasive beavers *Castor canadensis* increase patch-level avian diversity in Tierra del Fuego. *Journal of Applied Ecology*. doi.org/10.1111/1365-2664.13999 (IF=6.53)

    Contribution: Provided feedback on the analysis, edited the manuscript.

33. Fahey, R.T., Tanzer, D., Alveshere, B., Atkins, J.W., Gough, C.M., Hardiman, B.S. An experimental approach for crown to whole-canopy defoliation in forests. *Canadian Journal of Forest Research.* doi.org/10.1139/cjfr-2020-0527 (IF=1.82)

    Contribution: Co-conceived the research question, assisted with field data collection, and edited the manuscript.

32. Gough, C.M. Bohrer, G., Hardiman, B.S., Nave, L.E., Vogel, C.S., Atkins, J.W., Bond-Lamberty, B., Fahey, R.T., Fotis, A.T. Grigri, M.S., Haber, L.T., Ju, Y., Kleinke, C.L., Mathes, K.C., Nadelhoffer, K.J., Stuart-Haëntjens, E., Curtis, P.S. Disturbance-accelerated succession increases the production of a temperate forest. 2021. *Ecological Applications*. doi.org/10.1002/eap.2417 (IF=4.5)

    Contribution: Collected and analyzed data, and edited the manuscript.

31. LaRue, E.A.#, Dodds, W.K., Rohr, J., Dahlin, K., Thorp, J.H., Johnson, J.S., Hardiman, B.S., Rodriguez-Gonzalez, M.I.†, Keller, M.§, Fahey, R.T., Knott, J., SanClements, M., Atkins, J.W.#, Tromboni, F., Chandra, S., Parker, G., Rose, K., Liu, J., Fei, S. The evolution of macrosystems biology. 2021. *Frontiers in Ecology and the Environment*. doi:10.1002/fee.2288 (Invited) (IF=10.94)

Contribution: Wrote sections of the text, edited the manuscript.

30. Gough, C.M., Atkins, J.W.#, Fahey, R.T., Hardiman, B.S., and E LaRue, E.A.#. Community and structural constraints on the complexity of eastern North American forests. 2020. *Global Ecology and Biogeography*. doi.org/10.1111/geb.13180 (IF=6.45)

    Contribution: Co-concieved the idea, provided feedback on the analysis, edited the manuscript.

29. Atkins, J.W.#, Bond-Lamberty, B. Fahey, R.T., Haber, L.T., Stuart-Haëntjens, E., Hardiman, B.S., LaRue, E.A., McNeil, B.E., Orwig, D.A., Stovall, A.E.L., and others. Application of multidimensional structural characterization to detect and describe moderate forest disturbance. 2020. *Ecosphere*. doi.org/10.1002/ecs2.3156 (IF=2.88)

    Contribution: Co-concieved the idea, provided feedback on the analysis, edited the manuscript.

28. Haber, L.T.†, Fahey, R.T., Wales, S.B., Pascuas, N.C., Currie, W.S., Hardiman, B.S., Gough, C.M. Forest Structure and Primary Production In Relation To Disturbance Severity. 2020. *Ecology and Evolution*. doi.org/10.1002/ece3.6209 (IF=2.34)

    Contribution: Contributed results of a statistical analyses, wrote sections of the text, edited the manuscript.

27. LaRue, E.A.#, Wagner, F.W.†, Fei, S., Atkins, J.W.#, Fahey, R.T., Gough, C.M., and Hardiman, B.S. Compatibility of aerial and terrestrial LiDAR for quantifying forest structural diversity. 2020. *Remote Sensing*. doi.org/10.3390/rs12091407 (IF=4.51)

    Contribution: Senior author. Advised and mentored co-first authors LaRue and Wagner. Co-concieved the idea, provided feedback on the analysis, edited the manuscript.

26. Reynolds, H.L., Brandt, L., Fischer, B.C., Hardiman, B.S., Moxley, D.J., Sandweiss, E., Speer, J., Fei, S. Implications of climate change for managing urban green infrastructure: an Indiana, U.S. case study. 2020. *Climatic Change*. doi.org/10.1007/s10584-019-02617-0 (IF=3.54)

    Contribution: Wrote sections of the text, edited the manuscript.

25. LaRue, E.A.#, Hardiman, B.S., Elliot, J.M. §, Fei, S. Structural diversity as a predictor of ecosystem function. 2019. *Environmental Research Letters.* doi.org/10.1088/1748-9326/ab49bb (IF = 6.42)

    Contribution: Co-concieved the idea, provided feedback on the analysis, edited the manuscript.

24. Fahey, R.T., Atkins, J.W.#, Gough, C.M., Hardiman, B.S., Nave, L.E., Tallant, J., Nadelhoffer, K.J., Vogel, C.S., Scheuermann, C., Stuart-Haentjens, E., Haber, L., Fotis, A., Ricart, R., Curtis, P.S. Defining a spectrum of integrative trait-based vegetation canopy structural types. 2019. *Ecology Letters*. doi.org/10.1111/ele.13388 (IF=9.14)

    Contribution: Co-conceived the research question, helped design the research, assisted with data analysis, and edited the manuscript.

23. Gough, C.M., Atkins, J.W. #, Fahey, R.T., Hardiman, B.S. High rates of primary production in structurally complex forests. 2019. *Ecology.* doi.org/10.1002/ecy.2864 (IF=4.81)

Contribution: Co-conceived the research question, helped design the research, assisted with data analysis, and edited the manuscript.

22. Hardiman, B. S.; LaRue, E. A.#; Atkins, J. W.#; Fahey, R. T.; Wagner, F. W. †; Gough, C. M. Spatial variation in canopy structure across forest landscapes. 2018. Published in a special issue of *Forests*. doi.org/10.3390/f9080474 (Invited) (IF=2.25)

    Contribution: Conceived the research question, designed the research, collected and analyzed data, and led the writing and editing of the manuscript.

21. LaRue, E.#; Atkins, J.#; Dahlin, K.; Fahey, R.; Fei, S.; Gough, C.; Hardiman, B.S. Linking Landsat to terrestrial LIDAR: Vegetation metrics of forest greenness are correlated with canopy structural complexity. 2018. *International Journal of Applied Earth Observation and Geoinformation*. doi.org/10.1016/j.jag.2018.07.001 (IF=4.36)

    Contribution: Senior author. Conceived the research question, helped design the research, assisted with data analysis, and edited the manuscript.

20. Atkins, J.W.#, Bohrer, G., Fahey, R., Hardiman, B.S., Morin, T., Stovall, A., Zimmerman, N., Gough, C.M. Quantifying forest and canopy structural complexity metrics from terrestrial LiDAR data using the forestr R package. 2018. *Methods in Ecology and Evolution*. doi.org/10.1111/2041-210X.13061 (IF=6.36)

    Contribution: Helped conceive the research question and design, assisted with derivation of metrics, assisted with development of code in published software, and edited the manuscript.

19. Sargent, M., Barrera, Y., Nehrkorn, T., Hutyra, L.R., Gately, C.K., Jones, T., McKain, K., Sweeney, C., Hegarty, J., Hardiman, B.S., Wofsy, S.C. Anthropogenic and biogenic CO_2 fluxes in the Boston urban region. 2018. *Proceedings of the National Academy of Science*. doi.org/10.1073/pnas.1803715115 (IF=10.4)

    Contribution: Provided data, edited the manuscript.

18. Atkins, J.W.#, Fahey, R., Hardiman, B.S., and Gough, C.M. Forest structural complexity predicts canopy light absorption at the sub-continental scale. 2018. *Journal of Geophysical Research - Biogeosciences*. doi.org/10.1002/2017JG004256 (IF=3.39)

    Contribution: Conceived the research question, helped design the research, assisted with data analysis, and edited the manuscript.

17. Fahey, R., Alveshere, B., Burton, J., D'Amato, A., Dickinson, Y., Keeton, W., Kern, C., Larson, A., Palik, B., Puettmann, K., Saunders, M., Webster, C., Atkins, J.#, Gough, C., Hardiman, B.S. 2018. Shifting conceptions of complexity in forest management and silviculture. *Forest Ecology and Management*. doi.org/10.1016/j.foreco.2018.01.011 (IF=3.39)

    Contribution: Co-conceived the research question and edited the manuscript.

16. Fotis, A.T., Morin, T.H., Fahey, R.T., Hardiman B.S., et al. Forest structure in space and time: Biotic and abiotic determinants of canopy complexity and their effects on net primary productivity. 2018. *Agricultural and Forest Meteorology*. doi.org/10.1016/j.agrformet.2017.12.251 (IF=4.75)

    Contribution: Co-conceived the research question, helped design the research, and edited the manuscript.

15. Hardiman, B.S., Gough, C.M., Butnor, J.R., Bohrer, G., Detto, M., Curtis, P.S. Coupling Fine-Scale Root and Canopy Structure Using Ground-Based Remote Sensing. 2017. *Remote Sensing.* doi:10.3390/rs9020182 *(IF=3.75)*

14. Hardiman, B.S., Wang, J., Hutyra, L.R., Gately, C., Getson, J., Friedl, M. Accounting for urban biogenic fluxes in regional carbon budgets. 2017. *Science of the Total Environment* doi:10.1016/j.scitotenv.2017.03.028 (IF=5.10)

13. Gough, C.M., Curtis, P.S., Hardiman, B.S., Scheuermann, C. and Bond-Lamberty, B. Disturbance, complexity, and succession of net ecosystem production in North America's temperate deciduous forests. 2016. *Ecosphere*. doi.org/10.1002/ecs2.1375 (IF=2.29)

    Contribution: Co-conceived the research question, helped design the research, assisted with data analysis, and edited the manuscript.

12. Nguyen, H,T., Hutyra, L.R., Hardiman, B.S., Raciti, S.M. Characterizing Forest Structure Variations Across an Intact Tropical Peat Dome Using Field Samplings and Airborne Lidar. 2016. *Ecological Applications*. doi.org/10.1890/15-0017 (IF=4.31)

    Contribution: Helped with data analysis and edited the manuscript.

11. Bryan, A.M., Cheng, S.J., Ashworth, K., Guenther, A.B., Hardiman, B.S., Bohrer, G. and Steiner, A.L. Forest-atmosphere BVOC exchange in diverse and structurally complex canopies: 1-D modeling of a mid-successional forest in northern Michigan. 2015. *Atmospheric Environment*. http://dx.doi.org/10.1016/j.atmosenv.2015.08.094 (IF=3.95)

    Contribution: Collected and analyzed data, and edited the manuscript.

10. Viskari, T.T., Hardiman, B.S., Desai, A.R., and Dietze, M.C. Model-data assimilation of multiple phenological observations to constrain and predict leaf area index. 2015. *Ecological Applications*. doi.org/10.1890/14-0497.1 (IF=4.31)

    Contribution: Helped with data analysis and edited the manuscript.

9. Nave, L.E., Sparks, J.P., Le Moine, J., Hardiman, B.S., Nadelhoffer, K.J., Tallant, J.M., Vogel, C.S., Strahm, B.D., and Curtis, P.S. Changes in soil nitrogen cycling in a northern temperate forest ecosystem during succession. 2014. *Biogeochemistry*. doi.org/10.1007/s10533-014-0013-z (IF=3.43)

    Contribution: Helped design the research, assisted with data collection and analysis, and edited the manuscript.

8. Maurer, K.D., Hardiman, B.S., Vogel, C.S. and Bohrer, G. Canopy-structure effects on surface roughness parameters: Observations in a Great Lakes mixed-deciduous forest. 2013. *Agricultural and Forest Meteorology*. doi.org/10.1016/j.agrformet.2013.04.002 (IF=4.75)

    Contribution: Collected and analyzed data, and edited the manuscript.

7. Hardiman, B.S., Bohrer, G., Gough, C.M., and Curtis, P.S. Canopy Structural Changes Following Widespread Mortality of Canopy Dominant Trees. 2013. *Forests*. doi:10.3390/f4030537 (IF=2.25)

6. Hardiman, B.S., Gough, C.M., Halperin, A.§, Hofmeister, K.L.§, Nave, L.E., Bohrer, G. and Curtis, P.S. Maintaining high rates of carbon storage in old forests: A mechanism linking canopy structure to forest function. 2013. *Forest Ecology and Management*.

http://dx.doi.org/10.1016/j.foreco.2013.02.031 (IF=3.39)

5. Gough, C.M., Hardiman, B.S., Nave, L.E., Bohrer, G., Maurer, K.D., Vogel, C.S., Nadelhoffer, K.J. and Curtis, P.S. Sustained carbon uptake and storage following moderate disturbance in a Great Lakes forest. 2013. *Ecological Applications*. doi.org/10.1890/12-1554.1 (IF=4.31)

   Contribution: Conceived the research question, helped design the research, assisted with data collection and analysis, and edited the manuscript.

4. Garrity, S.R., Meyer, K.§, Maurer, K.D., Hardiman, B.S. and Bohrer, G. Estimating plot-level tree structure in a deciduous forest by combining allometric equations, spatial wavelet analysis and airborne LiDAR. 2012. *Remote Sensing Letters*. doi.org/10.1080/01431161.2011.618814 (IF=1.53)

   Contribution: Co-conceived the research question, collected and analyzed data, and edited the manuscript.

3. Nave, L.E., Gough, C.M., Maurer, K.D., Bohrer, G., Hardiman, B.S., Le Moine J., Munoz, A.B.§, Nadelhoffer, K.J., Sparks, J.P., Strahm, B.D., Vogel, C.S., and Curtis, P.S. Disturbance and the resilience of coupled carbon and nitrogen cycling in a north temperate forest. 2011. *Journal of Geophysical Research - Biogeosciences*. doi.org/10.1029/2011JG001758 (IF=3.40)

   Contribution: Helped collect and analyze the data, and edited the manuscript.

2. Hardiman, B.S., Bohrer, G., Gough, C.M., Vogel, C.S., and Curtis, P.S. The role of canopy structural complexity in wood net primary production of a maturing northern deciduous forest. 2011. *Ecology*. doi.org/10.1890/10-2192.1 (IF=4.81)

1. Gough, C.M., Vogel, C.S., Hardiman, B.S., and Curtis, P.S. Wood net primary production resilience in an unmanaged forest transitioning from early to middle succession. 2010. *Forest Ecology and Management*. doi.org/10.1016/j.foreco.2010.03.027 (IF=3.39)

   Contribution: Collected and analyzed data, and edited the manuscript.

Submitted manuscripts (N=5 as of 9/29/2021).

5. Hardiman, B.S., Tielens, E., Young, A., Song, Y., Chandrasekaran, A., LaRue, E., Atkins, J., Murray, B., Shao, G., McNickle, G., Dahlin, K., Chen, X. Observing and measuring forest vertical structure across spatial scales. In review at *Frontiers in Ecology and the Environment*. (IF=9.3)

4. LaRue, E., Fahey, R., Alveshere, B., Atkins, J., Bhatt, P., Buma, B., Chen, A., Cousins, S., Elliott, J., Elmore, A., Hakkenberg, C., Hardiman, B., Johnson, J., Kashian, D., Koirala, A ., Papeş, M., St. Hilaire, J., Surasinghe, T., Zambrano, J., Zhai, L., Fei, S. The ecological role of structural diversity. In review at *Frontiers in Ecology and the Environment*. (IF=9.3)

   Contribution: Co-conceived the research question and edited the manuscript.

3. Record, S., Jarzyna, M.A., Hardiman, B.S., Richardson, A.D. Interdisciplinary science and open data contribute to resilient innovation during pandemic. In review at *Frontiers in Ecology and the Environment*. (IF=9.3)

Contribution: Co-conceived the research question and edited the manuscript.

2. Rodriguez Gonzalez, M.I. †, Pijanowski, B.C., Fahey, R.T., Hardiman, B.S. Distribution and access of top-ranked ecosystem services in the Chicago metropolitan region. In review at *Landscape and Urban Planning*. (IF=5.44)

   Contribution: Co-conceived the research question, mentored the graduate student, and edited the manuscript.

1. LaRue, E.A.#, Dew, L.A., Hoyer, M., Hardiman, B.S. Herbicide control of invasive aquatic plants: ecological impacts and the need for a large-scale monitoring effort. In review at *Lake and Reservoir Management*. (IF=1.11)

   Contribution: Edited the manuscript.

Funding

Current externally funded research projects

1. Agency/Title of Grant: National Science Foundation. *Collaborative Research: MSA: Incorporating canopy structural complexity to improve model forecasts of functional effects of forest disturbance.*
2. Duration of Funding (Dates): 2019-2021
3. Total amount of award: $300,000
4. Role: PI
5. If co-PI, amount of the total funding for which I am directly responsible:

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1. Agency/Title of Grant: National Science Foundation. *MSB-FRA Modeling Invasion Dynamics Across Scales (MIDAS)*.
2. Duration of Funding (Dates): 2016-2021
3. Total amount of award: $996,171
4. Role: Senior personnel
5. If co-PI, amount of the total funding for which I am directly responsible: $50,000

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1. Agency/Title of Grant: National Science Foundation. *Macrosystems Biology and NEON enabled science investigator meeting*
2. Duration of Funding (Dates): 2020-2021
3. Total amount of award: $93,481
4. Role: Senior personnel
5. If co-PI, amount of the total funding for which I am directly responsible: $0

Current externally funded research projects

1. Agency/Title of Grant: Purdue Hardwood Tree Improvement and Regeneration Center (HTIRC). *Productivity-diversity Relationships in Hardwood Plantations*.
2. Duration of Funding (Dates): 2019-2021
3. Total amount of award: $ 150,298

4. Role: co-PI
5. If co-PI, amount of the total funding for which I am directly responsible: $1,200

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1. Agency/Title of Grant: Purdue Hardwood Tree Improvement and Regeneration Center (HTIRC). *Integrated digital forestry (iDiF): Using terrestrial laser scanning to assess tree health and quality*.
2. Duration of Funding (Dates): 2019-2021
3. Total amount of award: $149,694
4. Role: PI
5. If co-PI, amount of the total funding for which I am directly responsible:

Pending externally funded research projects

1. Agency/Title of Grant: United States Department of Agriculture: National Institutes of Food and Agriculture – Sustainable Agriculture Systems. *Promoting Economic Resilience and Sustainability of the Eastern US Forests (PERSEUS)*
2. Duration of Funding (Dates): 2022-2027
3. Total amount of award: $10,694,967
4. Role: co-PI
5. If co-PI, amount of the total funding for which I am directly responsible: $150,000

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1. Agency/Title of Grant: National Science Foundation. *Building community science networks with environmental sensors and online resources*
2. Duration of Funding (Dates): 2021-2024
3. Total amount of award: $693,266
4. Role: co-PI
5. If co-PI, amount of the total funding for which I am directly responsible: $173,316

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1. Agency/Title of Grant: National Science Foundation. *Teaching the Ecology of Technology and Big Data Management*
2. Duration of Funding (Dates): 2021-2024
3. Total amount of award: $608,628
4. Role: co-PI
5. If co-PI, amount of the total funding for which I am directly responsible: 152,157

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1. Agency/Title of Grant: National Science Foundation. Macrosystems Small Award: Determining mechanisms of coexistence for long-lived tree species across the USA
2. Duration of Funding (Dates): 2021-2024
3. Total amount of award: $289,068
4. Role: co-PI
5. If co-PI, amount of the total funding for which I am directly responsible: $72,267

1. Agency/Title of Grant: National Science Foundation. *Assessing the role of forest structural diversity in mediating continental-scale biodiversity and ecosystem multifunctionality*
2. Duration of Funding (Dates): 2020-2025
3. Total amount of award: $500,000
4. Role: co-PI
5. If co-PI, amount of the total funding for which I am directly responsible: $106,751

Pending internally funded research projects

1. Agency/Title of Grant: FNR Equipment Small Grants. *Full Spectrum Acoustic Wildlife Detector and Acoustic Lure Request*
2. Duration of Funding (Dates): 2021
3. Total amount of award: $5,374
4. Role: co-PI
5. If co-PI, amount of the total funding for which I am directly responsible: $776

Past externally funded research projects

1. Agency/Title of Grant: National Science Foundation. *Exploring new dimensions of forest ecosystems with structural diversity.*
2. Duration of Funding (Dates): 2019-2020
3. Total amount of award: $94,000
4. Role: co-PI
5. If co-PI, amount of the total funding for which I am directly responsible: $47,000

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1. Agency/Title of Grant: Indiana Water Resources Research Center. *Impacts of urban ecosystem services on human health and water quantity and quality in Northwestern Indiana communities: An unexplored opportunity to study service accessibility.*
2. Duration of Funding (Dates): 2019-2020
3. Total amount of award: $15,000
4. Role: PI
5. If co-PI, amount of the total funding for which I am directly responsible:

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1. Agency/Title of Grant: The American Chestnut Foundation. *Below ground competition of American chestnut*.
2. Duration of Funding (Dates): 2018-2019
3. Total amount of award: $9,950
4. Role: co-PI
5. If co-PI, amount of the total funding for which I am directly responsible: $1,850

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1. Agency/Title of Grant: National Science Foundation. *Collaborative Research: EAGER-NEON: Is Canopy Structural Complexity a Global Predictor of Primary Production?: Using NEON to Transform Understanding of Forest Structure-function.*
2. Duration of Funding (Dates): 2015-2018
3. Total amount of award: $300,000

4. Role: co-PI
5. If co-PI, amount of the total funding for which I am directly responsible: $59,952

Past internally funded research projects

1. Agency/Title of Grant: Purdue College of Engineering: Engineering Faculty Conversation (EFC) on Smart Cities. *Socio-ecological resilience of urban ecosystems to extreme climate events*.
2. Duration of Funding (Dates): 2019-2020
3. Total amount of award: $66,680
4. Role: co-PI
5. If co-PI, amount of the total funding for which I am directly responsible: $14,000

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1. Agency/Title of Grant: Purdue Climate Change Research Center. *Biomass remote sensing using P-band signals of opportunity: proof of concept*.
2. Duration of Funding (Dates): 2018-2019
3. Total amount of award: $7,000
4. Role: co-PI
5. If co-PI, amount of the total funding for which I am directly responsible: $500

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1. Agency/Title of Grant: Purdue College of Agriculture AgSEED. *Indiana hardwood forest structure and function: Seeking a functional mechanistic basis for species diversity-productivity relationships.*
2. Duration of Funding (Dates): 2017-2018
3. Total amount of award: $49,338
4. Role: co-PI
5. If co-PI, amount of the total funding for which I am directly responsible: $22,250

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1. Agency/Title of Grant: FNR Equipment Small Grants. *A shared library of environmental sensors.*
2. Duration of Funding (Dates): 2017
3. Total amount of award: $9,650
4. Role: PI
5. If co-PI, amount of the total funding for which I am directly responsible:

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1. Agency/Title of Grant: Purdue Mary S. Rice Farm Fund Equipment Grant. *Collaborative Research: Competition, coexistence and community structure: identifying the mechanisms that structure Indiana forests.*
2. Duration of Funding (Dates): 2016
3. Total amount of award: $10,000
4. Role: PI

5. If co-PI, amount of the total funding for which I am directly responsible:

Funding Awarded during Graduate School

2013 **ALOS PALSAR Award** (Alaska Satellite Facility) Award from ASF in the form of 643 scenes of satellite radar backscatter data from 21 forest research sites spanning eastern North America and Puerto Rico.

2012 Student Travel Award (**Department of Energy**), $1000 to cover travel and lodging while presenting research at the 2012 Terrestrial Ecosystem Science Principle Investigators Meeting.

2007-11 **NSF-IGERT Biosphere-Atmosphere Research and Training (BART),** University of Michigan Biological Station, (2-year fellowship and 2 merit-based summer awards) $60,000 for 24 month graduate stipend, $2,000 for research supplies, $1,000 for travel to 2 national conferences. $4600 for extension awards.

2010 **Janice Carson Beatley Fund (**OSU-EEOB) $670 for research supplies.

2010 **Henry Allan Gleason Fellowship** (University of Michigan Biological Station) $2,300 for 8 weeks of housing, board, and research fees at research station.

2009 **NCALM Seed Award** (National Center for Airborne Laser Mapping) Award from NCALM in the form of an aerial LiDAR survey of 40km^2 of land covering my entire research site in northern Michigan.

2006-07 **University Fellowship** (OSU) 11-month graduate fellow stipend.

Selected Presentations

Invited research presentations (N=10)

10. University of Wisconsin-Madison, Department of Atmospheric and Oceanic Sciences. “Modelling forest canopy structural and functional responses to disturbance.” 2021
9. Ashland University, Department of Biology. “Understanding the influence of forest canopy structure on ecosystem functions at continental scales.” 2020.
8. The Ohio State University, Department of Civil, Environmental, and Geodetic Engineering. “The influence of canopy structure on forest ecosystem function.” 2018.
7. Indiana Department of Natural Resources. “The future of carbon cycling in Indiana forests.” 2018.
6. The Ohio State University, Department of Evolution, Ecology & Organismal Biology. “Disturbing Ecosystems: Characterizing structure and function of ecosystems spanning a gradient of anthropogenic influence.” 2016.
5. Boston University Biogeosciences Interdisciplinary Seminar Series. “Canopy structure, disturbance, and the resilience of carbon storage in aging forests.” 2014.
4. Harvard Forest. “Resilience of Forest Carbon Storage through Disturbance and Succession.” 2013.
3. Eastern Michigan University, Department of Biology. “Increasing resource use efficiency maintains productivity of an aging northern deciduous forest.” 2011.
2. Ashland University, Department of Biology and Toxicology. “The role of disturbance and

succession in forest carbon storage." 2011.

1. University of Michigan Biological Station. External Advisory Committee Annual Meeting. "Succession and resource redistribution in a northern hardwood forest." 2009.

Keynote and Plenary Speaker Presentations (N=1)

1. 21st Central Hardwood Forest Conference. Indianapolis, IN. "21st century forestry: The promise and perils of emerging technologies." 2018.

Invited National and International Conference Presentations (N=3)

3. Ecological Society of America Annual Meeting. Salt Lake City, UT. "A consensus definition of structural diversity and its utility as a predictor of ecosystem function." (Invited) 2020.
2. Ecological Society of America Annual Meeting. Louisville, KY. "Understanding the influence of forest canopy structure on ecosystem functions at continental scales." (Invited) 2019.
1. Ecological Society of America Annual Meeting. Ft. Lauderdale, FL. "Is canopy complexity a global predictor of forest growth?: Using NEON to understand ecosystem structure-production relationships across broad spatial scales." (Invited panelist) 2016.

In addition to the invited presentations enumerated above, Dr. Hardiman has presented as lead-author at 11 national or international conferences (7 since joining Purdue). Dr. Hardiman mentored postdocs who were lead authors on 9 presentations at national/international conferences (all since joining Purdue). He was primary advisor for graduate students who were lead authors on 7 oral presentations at university symposia (N=1) or national/international conferences (N=6) and 2 poster presentations at national (N=1) and international (N=1) conferences. Dr. Hardiman was primary advisor for undergraduate students who were lead authors on 5 presentations at university symposia.

Mentoring

Graduate students advised (current; N=2)

2. Kanaan Hardaway, Purdue University, ESE/EEE, PhD, 2020-Present.

   Dissertation: The contribution of vegetation to equitable resilience of urban systems under extreme climate events.

1. Lindsey Darling, Purdue University, FNR, PhD, 2020-Present.

   Dissertation: The historical origins of the midwestern urban canopy and implications for sustainable and equitable management of future urban forests.

Graduate students advised (graduated; N=5)

5. Benjamin McCallister, Purdue University, FNR, MS, 2019-2021.
   Thesis: Effects of fragmentation on structure and function of urban and rural forests.

4. Mayra I. Rodríguez González, Purdue University, FNR, PhD, 2016-2021.
   Dissertation: The spatial, biophysical and social factors driving community access to ecosystem services in the Chicago metropolitan region.
3. Taylor Blanford, Purdue University, EEE, MS (non-thesis), 2019- 2020.
   Project: The contribution of urban vegetation to resilience of built infrastructure at risk from extreme climate events.
2. Franklin Wagner, Purdue University, FNR, MS, 2017-2019. Co-advised by Songlin Fei (FNR). Thesis: Linking Airborne and Terrestrial LiDAR to Improve Modeling of Forest Structure and Function.
1. Jacob Klaybor, Purdue University, EEE, MS (non-thesis), 2018-2019.
   Project: Estimating the Biomass of the Indianapolis and Chicago Urban Forests

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Graduate student committees (N=6)

6. Dante Francomano, Purdue University, FNR, Ph.D. (2016 – 2020)
5. Jessica Outcalt, Purdue University, FNR, Ph.D. (2016 – 2020)
4. Kliffi Subida Blackstone, Botany, Ph.D. (2017 – present)
3. Jay Mendoza Tomlin, Chemistry, Ph.D. (2018 - present)
2. Sujan Dawadi, Entomology, Ph.D. (2018 – present)
1. Kyle Earnshaw, Purdue University, FNR, Ph.D. (2016 – 2017)

Undergraduate students supervised (completed; N=13)

15. Jack Lyall, Purdue University, EEE major, (2021-current)
14. Julie McAuly, Purdue University, EEE major, (2021-current)
13. Breanna Motsenbocker, Purdue University, EEE major, (2019 – 2021)
    2020 Summer Undergraduate Research Fellowship (SURF);
    *Breanna is currently finishing a first-authored publication based on this work for submission to the *American Journal of Undergraduate Research* in Fall 2021.
12. Serae Neidigh, Purdue University, Honors NRES major, (January - May 2021)
11. Jamille St. Hilaire, Purdue University, EEE major, (2019 – 2020)
10. Gillian Clark, Purdue University, EEE major, BS (2016 – 2020)
9. Kellee Edington, Purdue University, FNR major, BS (2019 – 2020)
8. Taylor Blanford, Purdue University, NRES major, BS (2018-2019)
7. Jacob Klaybor, Purdue University, NRES major, BS (2017-2018)
6. Nate Ibarluzea, Purdue University, EEE major, BS (2018 – 2019)
5. Emily Summers, Purdue University, EEE major, BS (2018 – 2019)
4. Landon Neumann, Purdue University, FNR major, BS (2018 – 2019)
3. Yudi Li, Purdue University, Biology major, BS (2016 – 2017)
2. Xiangxing Long, Purdue University, EEE major, BS (2016 – 2018)
1. Kevin Hill, Purdue University, EEE major, BS (2016)
   *2016 Purdue Summer Undergraduate Research Fellowship (SURF)

Outreach & Engagement

Dr. Hardiman feels strongly that translation of research results for lay audiences and engagement with stakeholders is critically important. Outreach and engagement are integrated throughout Dr. Hardiman's research and teaching portfolios. Several examples are detailed below and in accompanying documents.

Conferences, meetings, and workshops organized

Dr. Hardiman has convened five sessions at national and international conferences. Each session provided a venue for students to present their graduate and undergraduate students to present the results of their research alongside professional scientists from around the world and receive critical feedback. At the 2014 and 2015 Fall Meetings of the American Geophysical Union, he co-convened oral and poster sessions on Ecosystem Structure: Remote Sensing Observations and Modeling of Its Influence on Radiation Regimes and Gas Exchanges that brought together leading scientists studying the influence of canopy structure on forest ecosystem function. In 2016, he was lead convener of a session on Advancing Understanding of Ecosystem Structure and Function through Remote Sensing. This session received 54 submitted abstracts, enough to be allocated the maximum of three oral sessions and a poster session. Dr. Hardiman served as the Outstanding Student Paper Award liaison for these sessions In this role, he identified, registered, and coordinated volunteer judges to evaluate and provide feedback on all student presentations (oral and poster). At the 2018 meeting of the Ecological Society of America, he convened a session on "Cutting-Edge Remote Sensing Applications in Ecology: Spanning Scales, Sensors, and Ecosystems". Dr. Hardiman organized a session at the 2019 Ecological Society of America meeting on "Novel Ecosystem Dynamics in Human Dominated Ecosystems." This session brought together a diverse and interdisciplinary group of leading scientists to address the complex interactions of humans with natural ecosystems.

In 2020, Dr. Hardiman served as a member of the organizing committee for the annual all-investigators meeting of the NSF Macrosystems Biology Program. This committee was tasked with determining the theme and schedule for the meeting, inviting keynote speakers, and running/facilitating two-day virtual conference for 180 participants in January 2021. This meeting was supported by a grant from NSF on which Dr. Hardiman was a co-PI. This conference also generated survey data that was the basis for a publication in review at *Frontiers in Ecology and the Environment*.

Also in 2020, Dr. Hardiman co-organized a NSF-supported workshop on the topic of *Forest Structural Diversity*. This two-day workshop convened 75 scientists from a wide array of disciplines to discuss topics and research questions related to forest structural diversity. A special issue proposal was submitted to *Frontiers in Ecology and the Environment* as an outcome of this workshop; the issue was approved and its constituent manuscripts are under review or revision currently.

Published code packages, data, and web tutorials (N=2)

2. Calculating Forest Structural Diversity Metrics from NEON LiDAR Data. (2020) https://www.neonscience.org/resources/learning-hub/tutorials/structural-diversity-discrete-return
Contribution: Hardiman is not a named author, but he advised and mentored first author LaRue, co-conceived the idea, and provided informal feedback on the tutorial. Hardiman

is senior author on the peer-reviewed publication that was the basis for this tutorial (See publication 29 in Section 1: Discovery).

1. Atkins, J.W., Bohrer, G., Fahey, R.T., Hardiman, B.S., Gough, C.M., Morin, T., Stovall, A., and Zimmerman, N. (2020). forestr: Ecosystem and Canopy Structural Complexity Metrics from LiDAR. R package version 2.0.2. https://CRAN.R-project.org/package=forestr

   Contribution: Authored code, advised the lead author.

Technical reports for non-specialist audiences (N=3)

3. Reynolds, H.L., Brandt, L., Fischer, B.C., Hardiman, B.S., Moxley, D.J., Sandweiss, E., Speer, J., Dukes, J.S. Maintaining Indiana's Urban Green Spaces: A Report from the Indiana Climate Change Impacts Assessment. 2018. Purdue Climate Change Research Center, Purdue University. West Lafayette, Indiana. https://docs.lib.purdue.edu/cgi/viewcontent.cgi?article=1000&context=urbantr

   Contribution: Wrote sections of the text, edited the report.

2. Reynolds, H.L., Brandt, L., Fischer, B.C., Hardiman, B.S., Moxley, D.J., Sandweiss, E., Speer, J., Fei, S. Implications of climate change for managing urban green infrastructure in Indiana. 2018. *Urban Green Spaces Publications*. https://docs.lib.purdue.edu/urbanpub/1/

   Contribution: Wrote sections of the text, edited the report.

1. Rodriguez-Gonzalez, M. I., Clark, G., Hardiman, B. S., Meilan, R., Sutherland, J. W. (2017) Planting trees to reduce atmospheric levels of CO2.

   Contribution: Wrote sections of the text, edited the report.

Presentations for practitioners and general audiences (N=4)

4. Data Driven Agriculture Webinar Series. "Digital Forestry Webinar: Key Developments and Applications in Terrestrial Remote Sensing." Invited Speaker. (2021)
3. Purdue Climate Change Research Center Virtual bag lunch: Climate solutions on agricultural and forested lands (Panel discussion). Panelist. (2020)
2. Indiana Department of Natural Resources Meeting of Professional Foresters. "The future of carbon cycling in Indiana forests." Keynote Speaker. (2018)
1. Indiana Association of Arborists annual meeting. "The critical role of vegetation in urban ecosystems." Invited Speaker. (2018)

Science Communications (N=1)

1. Interviewed by podcast host for Episode 38 of "Major Revisions: A podcast for early career ecologists." https://www.majorrevisionspodcast.com/episodes/ep-38-an-interview-with-brady-hardiman. (2018)

Service

Current and prior administrative service appointments (N=10)

2021-present	FNR Department Head Search Committee
2021-present	FNR Strategic Planning Task Team
2020-present	EEE Academics Committee

2020-present	FNR Diversity, Equity, and Inclusion Committee, Lead
2019-present	FNR Data Integrity and Quality (IQ) Task Team
2019-2021	College of Agriculture Grade Appeals Committee
2018-2019	FNR IOT Faculty Search Committee
2018-2020	EEE Graduate Committee
2017-2018	FNR Building Design Task Team
2016-present	FNR Awards Committee
2016-2018	EEE Academics Committee

National Organizing Committees:

Dr. Hardiman served as a member of the organizing committee for the 2021 All-PI meeting for the Macrosystems Biology program at NSF. The theme of this meeting was “Inclusive Team Science.” This meeting brought together scientists from around the country to share current research and to exchange ideas and best practices on building and sustaining collaborative teams that are diverse and inclusive. This meeting was funded by an award based on a proposal written by the organizing committee reviewed by program officers at The National Science Foundation.

Reviewer for national funding agencies (dates of service excluded to ensure confidentiality):

NSF: Ecosystem Science Panel
NSF: Sustainable Regional Systems Panel
NSF: Signals in the Soil (SitS) Panel

Reviewer for professional journals (N=17):

Agricultural and Forest Meteorology, Carbon Balance and Management, Ecology and Evolution, Ecological Complexity, Ecosystems, Forest Ecology and Management, Forests, Global Ecology and Biogeography, International Journal of Remote Sensing, Journal of Ecology, JGR-Atmospheres, JGR-Biogeosciences, Oecologia, PLoS ONE, Remote Sensing, Science of the Total Environment, TREES: Structure and Function

Professional Affiliations

American Geophysical Union
Ecological Society of America
American Association of Geographers
International Association of Landscape Ecologists